209651: metamonzogranite, Success Well

(Cundimurra Monzogranite, Tuckanarra Suite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Cue (SG 50-15), Austin (2442)
MGA Zone 50, 585333E 6911764N

Sampled on 10 October 2012

This sample was collected from a low outcrop on Wynyangoo Station, about 4.5 km west of Deep Well, 3.6 km northwest of Carey Well, and 3.1 km east-northeast of Success Well.

Geological context

The unit sampled is the Cundimurra Monzogranite of the Tuckanarra Suite (Van Kranendonk et al., 2013; Zibra et al., 2014; Zibra, 2020). The 2697–2657 Ma Cundimurra Monzogranite is a north-trending, elongate pluton, up to 200 km long and 25 km wide, which consists mainly of metamonzogranite and metasyenogranite, but also includes metatonalite, leucocratic muscovite metamonzogranite, and minor quartz metadiorite veins (Zibra, 2020). Plutons of the Tuckanarra Suite consist of strongly foliated and locally magmatically layered metagranodiorite to metamonzogranite. Tuckanarra Suite metagranitic rocks intrude the Murchison Supergroup and are crosscut by non-deformed granitic plutons of the Bald Rock Supersuite (Van Kranendonk et al. 2013; Zibra, 2020). The metamonzogranite at this locality is intruded by a younger metagranodiorite vein of the Cundimurra Monzogranite, dated at 2685 ± 6 Ma (GSWA 209650, Wingate et al., 2021). A monzogranite dyke, intruded into the c. 2724 Ma Crossroads Granodiorite (Zibra, 2019) about 47 km to the northwest, yielded an igneous crystallization age of 2684 ± 3 Ma and is assigned to the Tuckanarra Suite (GSWA 155514, Lu et al., 2017). Samples of the Cundimurra Monzogranite, collected about 130 and 144 km to the south, yielded igneous crystallization ages of 2679 ± 8 Ma (GSWA 185997, Wingate et al., 2015a) and 2681 ± 5 Ma (GSWA 198212, Wingate et al., 2015b), respectively.

Petrographic description

The sample is an altered, inequigranular, medium-grained biotite metamonzogranite (Fig. 1), consisting of about 45–50% plagioclase, 25–30% K-feldspar, 20–25% quartz, 2–3% altered biotite, and minor myrmekite, magnetite and zircon. Plagioclase (andesine, An_{38}) is subhedral, equant to elongate, up to 5 mm in size, and weakly to moderately altered to clay minerals, sericite, epidote, and muscovite. Many grains are compositionally zoned with altered cores and narrow transparent rims. K-feldspar forms anhedral, equant grains up to 6.5 mm in diameter, and much smaller grains between plagioclase laths. Most K-feldspar is strained and exhibits microcline twinning. The larger grains contain small plagioclase inclusions. Irregular patches of equant plagioclase and K-feldspar grains, up to 0.2 mm in size, display polygonal texture. Small lobes of myrmekite are rare. Quartz forms elongate aggregates up to 7 mm long of intergrown anhedral grains, which range up to 4 mm long, and are typically weakly to moderately strained. Chlorite-replaced biotite occurs in small clusters of subhedral laths to 0.5 mm in length. Subhedral magnetite up to 0.4 mm in size is partially replaced by hematite. The sample has a weak foliation defined by elongate quartz and micas, and shows evidence of deformation and some recovery at moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque and subhedral to euhedral. The crystals are up to 400 µm long, and mostly elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, many crystals contain high-U, metamict zones, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 13–14 August 2020, using SHRIMP-B. Ten analyses of the M257 standard were obtained during the session, of which eight analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.61% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.32% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

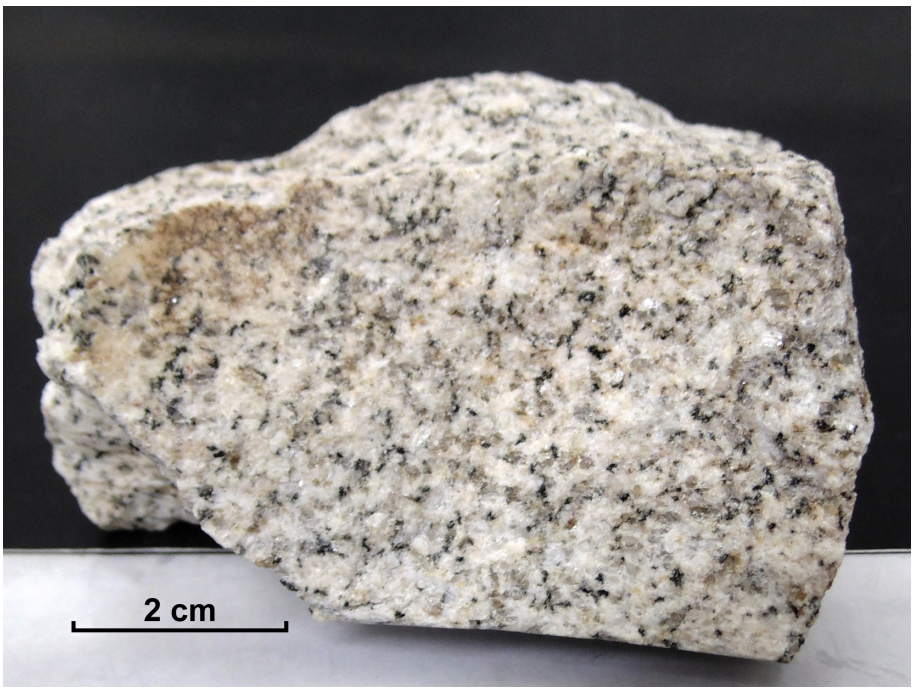

Figure 1. Hand specimen image of sample 209651: metamonzogranite, Success Well

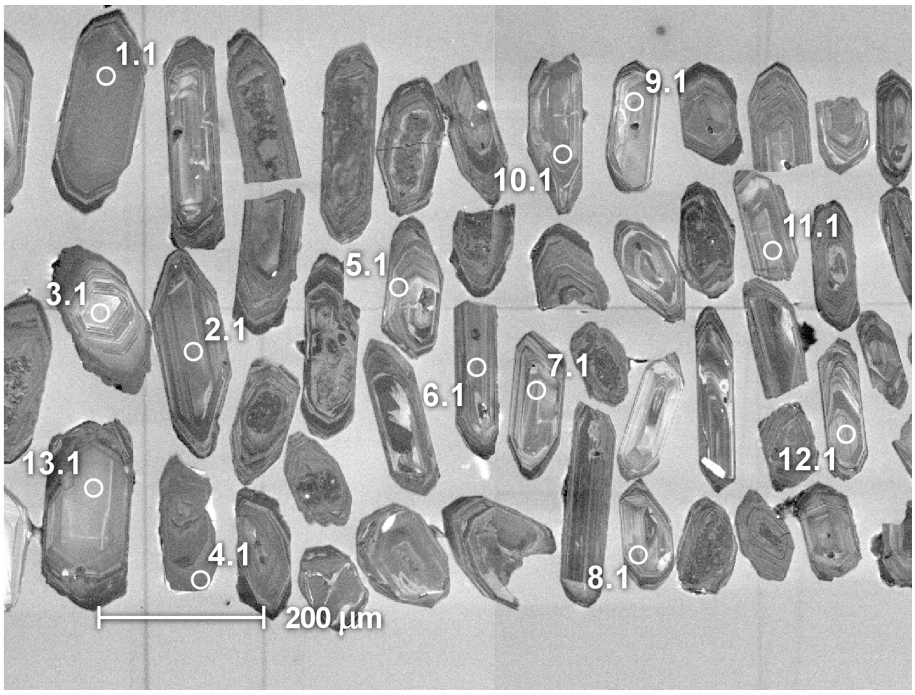

Figure 2. Cathodoluminescence image of representative zircons from sample 209651: metamonzogranite, Success Well. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 209651: metamonzogranite, Success Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	13	13.1	95	42	0.44	0.204	1.937	0.027	0.18347	0.00147	1.941	0.027	0.18165	0.00163	2679	31	2668	15	-0.4
I	4	4.1	336	121	0.36	0.024	2.105	0.020	0.18378	0.00076	2.106	0.020	0.18356	0.00077	2505	20	2685	7	6.7
I	10	10.1	260	257	0.99	0.011	1.934	0.020	0.18392	0.00088	1.934	0.020	0.18383	0.00088	2686	23	2688	8	0.1
I	16	16.1	119	87	0.73	0.093	1.918	0.025	0.18483	0.00130	1.920	0.026	0.18400	0.00137	2703	30	2689	12	-0.5
I	5	5.1	187	78	0.42	0.466	2.272	0.026	0.18819	0.00116	2.282	0.026	0.18403	0.00143	2343	23	2690	13	12.9
I	1	1.1	260	155	0.59	0.081	2.021	0.019	0.18543	0.00075	2.023	0.019	0.18470	0.00078	2589	21	2696	7	3.9
I	11	11.1	139	91	0.66	-0.019	1.941	0.024	0.18470	0.00116	1.941	0.024	0.18487	0.00118	2679	28	2697	11	0.7
I	6	6.1	204	213	1.04	0.245	2.017	0.022	0.18738	0.00101	2.022	0.022	0.18520	0.00114	2591	24	2700	10	4.0
I	18	18.1	255	235	0.92	0.126	2.034	0.021	0.18649	0.00089	2.036	0.021	0.18536	0.00095	2575	22	2702	8	4.7
I	7	7.1	160	123	0.77	0.285	2.331	0.123	0.18840	0.00124	2.337	0.123	0.18586	0.00141	2296	102	2706	13	15.2
I	9	9.1	161	51	0.31	0.372	2.070	0.054	0.18930	0.00114	2.078	0.055	0.18599	0.00134	2533	55	2707	12	6.4
I	12	12.1	104	87	0.83	0.000	1.954	0.026	0.18641	0.00140	1.954	0.026	0.18641	0.00140	2664	30	2711	12	1.7
I	8	8.1	163	123	0.76	0.265	2.031	0.023	0.18903	0.00112	2.037	0.023	0.18667	0.00126	2575	25	2713	11	5.1
I	17	17.1	155	49	0.31	-0.055	1.924	0.023	0.18713	0.00118	1.923	0.023	0.18763	0.00121	2699	27	2722	11	0.8
I	14	14.1	42	22	0.54	-0.067	1.904	0.038	0.18759	0.00223	1.903	0.038	0.18819	0.00231	2723	45	2726	20	0.1
I	2	2.1	317	219	0.69	0.175	2.059	0.080	0.19358	0.00668	2.063	0.080	0.19202	0.00670	2548	82	2760	57	7.7
X	15	15.1	116	55	0.47	0.157	1.907	0.025	0.20269	0.00140	1.910	0.025	0.20129	0.00150	2715	30	2837	12	4.3
P	3	3.1	173	38	0.22	0.000	2.982	0.063	0.11720	0.00102	2.982	0.063	0.11720	0.00102	1864	34	1914	16	2.6

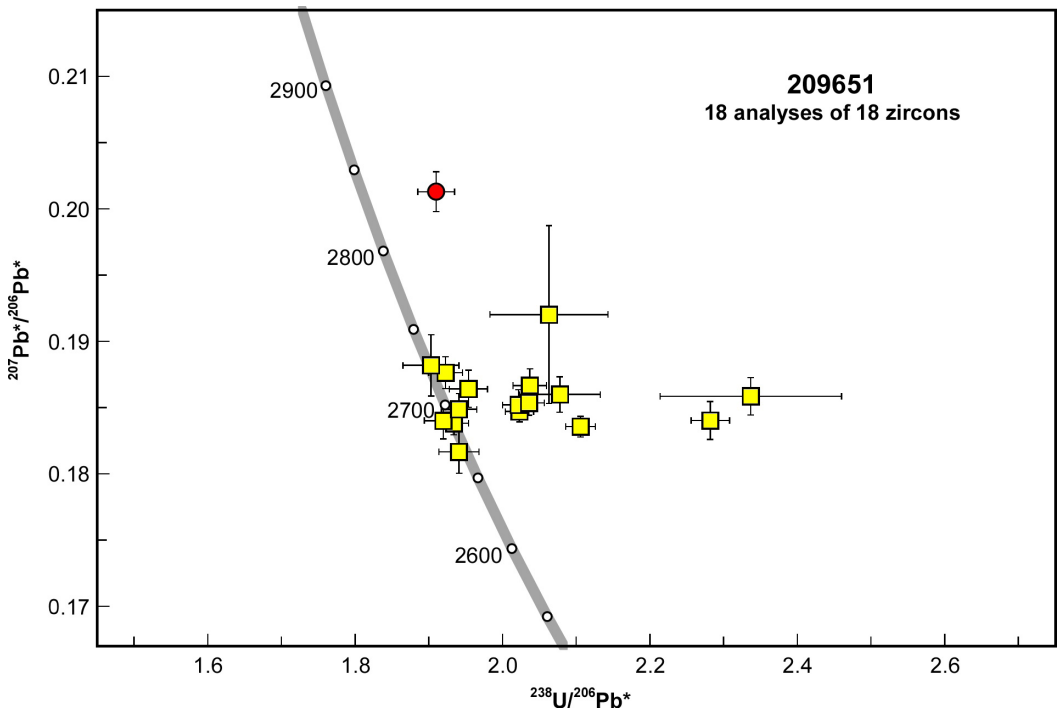

Figure 3. U–Pb analytical data for sample 209651: metamonzogranite, Success Well. Yellow squares indicate Group I (magmatic zircons); red circle indicates Group X (xenocrystic zircon). One analysis in Group P (radiogenic-Pb loss) is not shown

Interpretation

The analyses are concordant to strongly discordant (Fig. 3), and can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 16 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2698 ± 7 Ma (MSWD = 1.5).

Group X comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2837 ± 12 Ma (1σ).

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 1914 ± 16 Ma (1σ).

The date of 2698 ± 7 Ma for the 16 analyses in Group I is interpreted as the igneous crystallization age of the monzogranite. The date of 2837 ± 12 Ma (1σ) for the single analysis in Group X is interpreted as the age of an inherited zircon. The date of 1914 ± 16 Ma (1σ) for the single analysis in Group P is tentatively interpreted to reflect loss of radiogenic Pb.

References

Lu, Y, Wingate, MTD, Kirkland, CL and Zibra, I 2017, 155514: monzogranite dyke, Kowergabbie Well; Geochronology Record 1392: Geological Survey of Western Australia, 4p.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD, Kirkland, CL and Wyche, S 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

Wingate, MTD, Fielding, IOH, Lu, Y and Zibra, I 2021, 209650: metagranodiorite vein, Success Well; Geochronology Record 1830: Geological Survey of Western Australia, 4p.

Wingate, MTD, Kirkland, CL and Ivanic, TJ 2015a, 185997: granitic gneiss, Tandys Bore; Geochronology Record 1229: Geological Survey of Western Australia, 3p.

Wingate, MTD, Kirkland, CL and Ivanic, TJ 2015b, 198212: porphyritic monzogranite, Meroin Rock; Geochronology Record 198212: Geological Survey of Western Australia, 4p.

Zibra, I 2019, Crossroads Granodiorite (A-SDcr-mgg): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 16 November 2021, <www.dmp.wa.gov.au/ens>.

Zibra, I 2020, Cundimurra Monzogranite (A-TUcu-mg): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 16 November 2021, <www.dmp.wa.gov.au/ens>.

Zibra, I, Smithies, RH, Wingate, MTD and Kirkland, CL 2014, Incremental pluton emplacement during inclined transpression: Tectonophysics, v. 623, p. 100–122, doi:10.1016/j.tecto.2014.03.020.

Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Zibra, I 2021, 209651: metamonzogranite, Success Well; Geochronology Record 1831: Geological Survey of Western Australia, 4p.

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